

# **LAB SHEET**



**K.R. MANGALAM UNIVERSITY**

**SOET**

**(School of Engineering & Technology)**

**B.TECH CSE (CORE)**

**Database & Management System Lab (ENCS254)**

**Submitted by-**

Dolly

Roll no - 2401010196

**Submitted to-**

Mansi Kajal

## LAB SHEET - 3

### Objectives

- To understand the use of COMMIT and ROLLBACK commands in transaction control.
- To use ORDER BY and GROUP BY clauses in SQL queries.
- To understand DCL commands (GRANT, REVOKE) and the use of SAVEPOINT.
- To understand and implement JOIN operations in SQL.

### ○ Commands:

- • **TCL commands:** COMMIT, ROLLBACK, SAVEPOINT are used to control transactions.
- • **DCL commands:** GRANT and REVOKE are used to give or remove user privileges.
- • **GROUP BY** is used for aggregation.
- • **ORDER BY** is used for sorting.
- • **JOIN** is used to combine data from multiple tables.

### • DDL Commands

1. Create TEACHER Table

```
CREATE TABLE TEACHER (  
    Name VARCHAR(50),  
    DeptNo INT,  
    Date_of_Joining DATE,  
    DeptName VARCHAR(50),  
    Location VARCHAR(50),  
    Salary DECIMAL(10,2)  
);
```

2. Create SALES Table

```
CREATE TABLE SALES (  
    Sales_No INT PRIMARY KEY,  
    Salesname VARCHAR(50),  
    Branch VARCHAR(50),  
    Salesamount DECIMAL(10,2),  
    DOB DATE  
);
```

3. Create DEPARTMENTS Table

```

• CREATE TABLE DEPARTMENTS (
    dept_no INT PRIMARY KEY,
    dept_name VARCHAR(50),
    dept_location VARCHAR(50)
);

```

#### 4. Create DEPARTMENTS Table

```

• CREATE TABLE EMPLOYEES (
    emp_id INT PRIMARY KEY,
    emp_name VARCHAR(50),
    emp_salary DECIMAL(10,2),
    dept_no INT,
    FOREIGN KEY (dept_no) REFERENCES DEPARTMENTS(dept_no)
);

```

### • DML Operations

#### 1. Insert 5 Records

```

• INSERT INTO TEACHER VALUES
('Ravi',1,'2015-06-10','Mathematics','Delhi',50000),
('Anita',2,'2017-03-15','Commerce','Mumbai',45000),
('Karan',1,'2018-09-20','Mathematics','Delhi',55000),
('Neha',3,'2016-01-12','Science','Chandigarh',48000),
('Amit',2,'2019-11-05','Commerce','Mumbai',47000);

```

```

• INSERT INTO SALES VALUES
(1,'Rohan','Delhi',20000,'1999-12-21'),
(2,'Priya','Mumbai',15000,'2000-05-10'),
(3,'Aman','Delhi',18000,'1998-12-15'),
(4,'Neha','Chandigarh',22000,'1997-08-25'),
(5,'Karan','Mumbai',25000,'1996-12-05');

```

```

• INSERT INTO DEPARTMENTS VALUES
(1,'Mathematics','Delhi'),
(2,'Commerce','Mumbai'),
(3,'Science','Chandigarh');

```

- ```
INSERT INTO EMPLOYEES VALUES
(101,'Rohit',60000,1),
(102,'Priya',45000,2),
(103,'Aman',50000,1),
(104,'Neha',55000,3),
(105,'Karan',40000,2);
```

2. 25% Increment for Mathematics Department

- ```
UPDATE TEACHER
SET Salary = Salary + (Salary * 0.25)
WHERE DeptName = 'Mathematics';
```

3. 15% Increment for Commerce Department

- ```
UPDATE TEACHER
SET Salary = Salary + (Salary * 0.15)
WHERE DeptName = 'Commerce';
```

## • From SALES Table

1. Total Sales in Each Branch

- ```
SELECT Branch, SUM(Salesamount) AS Total_Sales
FROM SALES
GROUP BY Branch;
```

Output:

| Result Grid |            |             | Filter Rows: |
|-------------|------------|-------------|--------------|
|             | Branch     | Total_Sales |              |
| ▶           | Delhi      | 38000.00    |              |
|             | Mumbai     | 40000.00    |              |
|             | Chandigarh | 22000.00    |              |

2. Average Sales in Each Branch

- ```
SELECT Branch, AVG(Salesamount) AS Average_Sales
FROM SALES
GROUP BY Branch;
```

Output:

| Result Grid |            |               | Filter Rows: |
|-------------|------------|---------------|--------------|
|             | Branch     | Average_Sales |              |
| ▶           | Delhi      | 19000.000000  |              |
|             | Mumbai     | 20000.000000  |              |
|             | Chandigarh | 22000.000000  |              |

### 3. Salesname and DOB (Born in December)

```

• SELECT Salesname, DATE_FORMAT(DOB, '%d-%b-%y') AS DOB
FROM SALES
WHERE MONTH(DOB) = 12;

```

Output:

| Result Grid |           |           | Filter Rows: |
|-------------|-----------|-----------|--------------|
|             | Salesname | DOB       |              |
| ▶           | Rohan     | 21-Dec-99 |              |
|             | Aman      | 15-Dec-98 |              |
|             | Karan     | 05-Dec-96 |              |

### 4. Salesname and DOB in Alphabetical Order of Month

```

• SELECT Salesname, DOB
FROM SALES
ORDER BY MONTHNAME(DOB);

```

Output:

| Result Grid |           |            | Filter Rows: |
|-------------|-----------|------------|--------------|
|             | Salesname | DOB        |              |
| ▶           | Priya     | 2000-05-10 |              |
|             | Neha      | 1997-08-25 |              |
|             | Rohan     | 1999-12-21 |              |
|             | Aman      | 1998-12-15 |              |
|             | Karan     | 1996-12-05 |              |

## • JOIN Operations

### 1. Employee and Department Details (INNER JOIN)

| Result Grid  |        |          |            |             |               |
|--------------|--------|----------|------------|-------------|---------------|
| Filter Rows: |        |          |            |             |               |
|              | emp_id | emp_name | emp_salary | dept_name   | dept_location |
| ▶            | 101    | Rohit    | 60000.00   | Mathematics | Delhi         |
|              | 103    | Aman     | 50000.00   | Mathematics | Delhi         |
|              | 102    | Priya    | 45000.00   | Commerce    | Mumbai        |
|              | 105    | Karan    | 40000.00   | Commerce    | Mumbai        |
|              | 104    | Neha     | 55000.00   | Science     | Chandigarh    |

```

• SELECT *
  FROM EMPLOYEES E
 JOIN DEPARTMENTS D
  ON E.dept_no = D.dept_no;

```

Output:

| Result Grid  |          |             |
|--------------|----------|-------------|
| Filter Rows: |          |             |
|              | emp_name | dept_name   |
| ▶            | Rohit    | Mathematics |
|              | Priya    | Commerce    |
|              | Aman     | Mathematics |
|              | Neha     | Science     |
|              | Karan    | Commerce    |

## 2. LEFT OUTER JOIN

```

• SELECT E.emp_name, D.dept_name
  FROM EMPLOYEES E
 LEFT JOIN DEPARTMENTS D
  ON E.dept_no = D.dept_no;

```

Output:

| Result Grid  |          |             |
|--------------|----------|-------------|
| Filter Rows: |          |             |
|              | emp_name | dept_name   |
| ▶            | Rohit    | Mathematics |
|              | Aman     | Mathematics |
|              | Priya    | Commerce    |
|              | Karan    | Commerce    |
|              | Neha     | Science     |

## 3. RIGHT OUTER JOIN

- ```
SELECT E.emp_name, D.dept_name
FROM EMPLOYEES E
RIGHT JOIN DEPARTMENTS D
ON E.dept_no = D.dept_no;
```

| Result Grid |          | Filter Rows: |
|-------------|----------|--------------|
|             | emp_name | dept_name    |
| ▶           | Rohit    | Mathematics  |
|             | Aman     | Mathematics  |
|             | Priya    | Commerce     |
|             | Karan    | Commerce     |
|             | Neha     | Science      |

#### 4. Employees with Salary Greater than Average

- ```
SELECT *
FROM EMPLOYEES
WHERE emp_salary > (SELECT AVG(emp_salary) FROM EMPLOYEES);
```

Output:

Result Grid

Filter Rows:

|   | emp_id | emp_name | emp_salary | dept_no |
|---|--------|----------|------------|---------|
| ▶ | 101    | Rohit    | 60000.00   | 1       |
|   | 104    | Neha     | 55000.00   | 3       |
| * | NULL   | NULL     | NULL       | NULL    |

### • TCL Commands

1. ROLLBACK
2. COMMIT
3. SAVEPOINT

- ```
SAVEPOINT A;
```
- ```
ROLLBACK;
```
- ```
COMMIT;
```

### • DCL Commands

1. Grant All Privileges

- `CREATE USER IF NOT EXISTS 'dept_user'@'localhost' IDENTIFIED BY '1234';`
- `GRANT ALL PRIVILEGES ON DBMS_LAB3.EMPLOYEES TO 'dept_user'@'localhost';`

## 2. Grant Selected Privileges

- `GRANT SELECT, INSERT ON DBMS_LAB3.EMPLOYEES TO 'dept_user'@'localhost';`

## 3. Revoke All Privileges

- `REVOKE SELECT, INSERT ON DBMS_LAB3.EMPLOYEES FROM 'dept_user'@'localhost';`

## 4. Revoke Selected Privileges

```
REVOKE ALL PRIVILEGES ON DBMS_LAB3.EMPLOYEES FROM 'dept_user'@'localhost';
```